Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 32. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the f1 score during training.

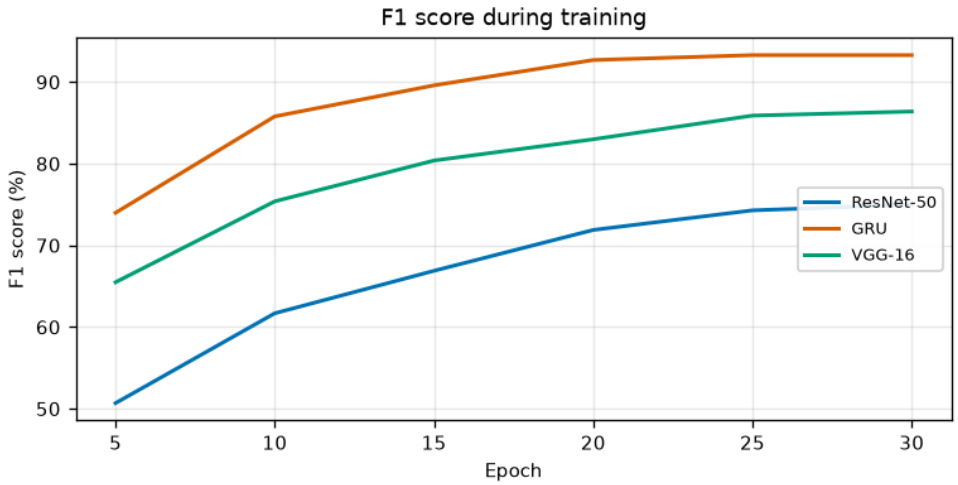


Figure 1: F1 score during training for the models in this experiment.

GRU outperforms ResNet-50 by 18 percentage points at epoch 30. GRU outperforms ResNet-50 by 23.3 percentage points at epoch 5 (Figure 1). As shown in Figure 1, from epoch 20 to epoch 25, the F1 score of VGG-16 improves by a relative 31.7%. From epoch 25 to epoch 30, the F1 score of ResNet-50 improves by a relative 7.2%.

Figure 2 shows the test accuracy during training.

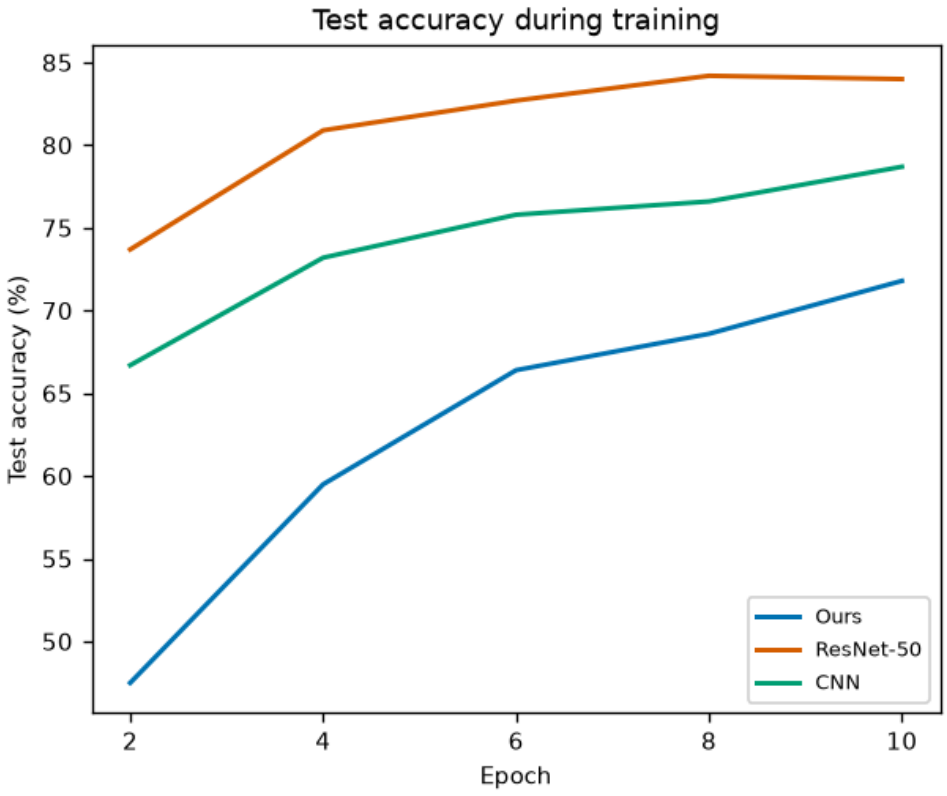


Figure 2: Test accuracy during training for the models in this experiment.

From epoch 4 to epoch 8, the test accuracy of Ours improves by a relative 15.3% (Figure 2). As shown in Figure 2, CNN attains a test accuracy of 77% at epoch 8.

Figure 3 shows the f1 score by dataset.

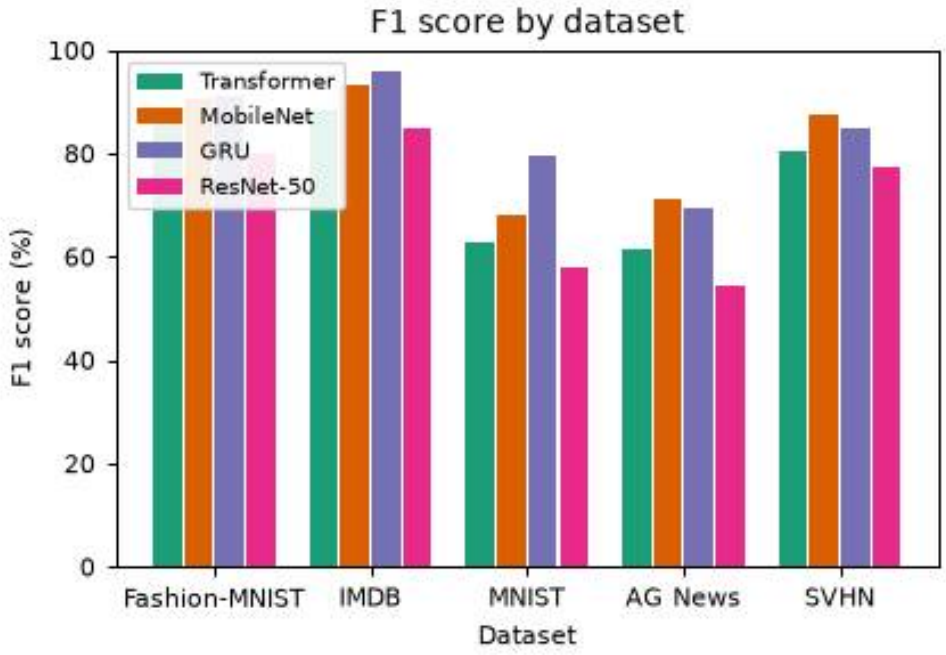


Figure 3: F1 score by dataset for the models in this experiment.

Transformer reaches 111.3% F1 score on AG News. Transformer achieves 90.5% F1 score on SVHN. As shown in Figure 3, GRU achieves 134.9% F1 score on SVHN. On AG News, MobileNet is 10 points better than Transformer (Figure 3).

Figure 4 shows the f1 score of ViT-S by dataset.

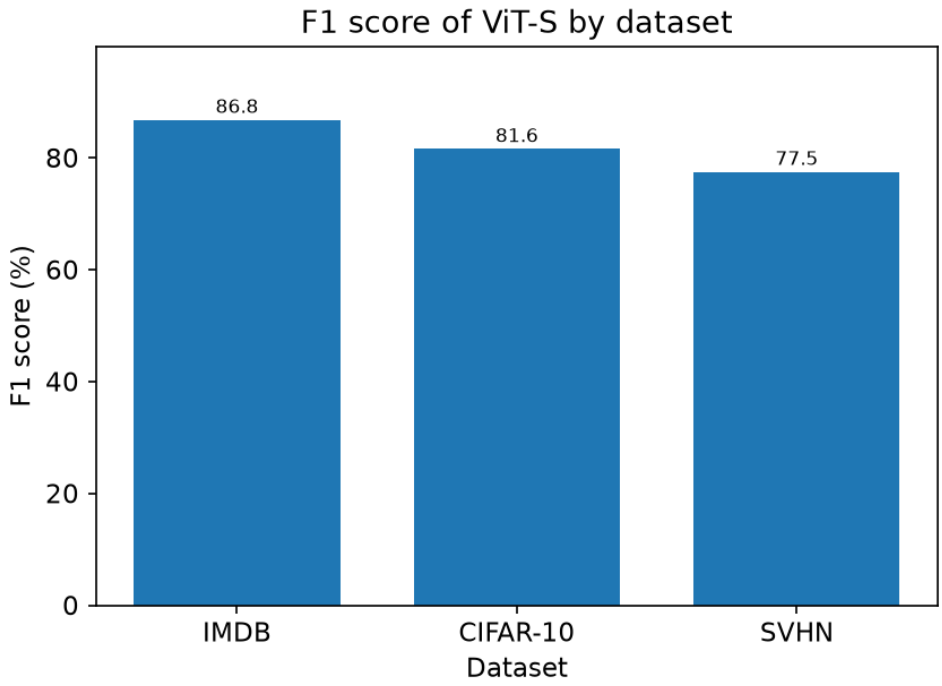


Figure 4: F1 score of ViT-S by dataset for the models in this experiment.

ViT-S scores 59.2 points higher on IMDB than on SVHN. ViT-S scores 9 points higher on IMDB than on

SVHN (Figure 4).

3 Discussion

On Fashion-MNIST, SVM obtains an accuracy of 87.8%. We note that the training was stable for all models.